ELECTRICAL MACHINES · CH3

Transformer equivalent circuit

Cheat sheet · Day 04 · Tue 27 Oct · pack M4

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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TRANSFORMER EQUIVALENT CIRCUIT (Ch3)                    n = N1/N2 (rated V1/V2)
Ideal:    V1 = nV2   I1 = I2/n   ZIN = n^2 ZL   S1 = S2
Refer to primary:  V2' = nV2   I2' = I2/n   Z' = n^2 Z   (angles unchanged)
          back:     V2 = V2'/n   I2 = n I2'
Simplified circuit (ref. to primary): RC || jXM across V1, then R + jX series to V2'
          R = R1 + R2' = R1 + n^2 R2      X = X1 + X2' = X1 + n^2 X2
TESTS (3ph star: V = VL/sqrt3, I = IL, P = Ptot/3 FIRST)   S = VI, Q = sqrt(S^2 - P^2
  OC (rated V, sec open)  -> PARALLEL:  RC = V^2/P    XM = V^2/Q
  SC (low V, sec shorted) -> SERIES:    R  = P/I^2    X  = Q/I^2     (check |Z| = V/I
  Params come out on the side you energised -> x n^2 to move LV -> HV
USING IT
  KVL:      V2' = V1 - I2'(R + jX)        V1 = V2' + I2'(R + jX)
  Divider:  V2' = V1 * ZL'/(ZL' + R + jX)
  I1 = I2' + V1/RC + V1/(jXM)
  Output conditions: V2' = V1rated(phase) <0 ;  I2' = S/V2' (1ph) or S/(sqrt3 VL) (3p
     angle phi = -acos(pf) lagging (absorbs Q), +acos(pf) leading (generates Q)
  Phase: resistive (or capacitive) load -> V2' LAGS V1 (series X makes I2' lag; V2' :
Reverse: no-load |I| = sqrt((V/RC)^2 + (V/XM)^2), P = V^2/RC ; SC: |I| = V/|R+jX|, P
Saturation: XM valid only at the test voltage (Prac 2: 3982 -> 366 ohm from 100 -> 24
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